

Roll No.

TCE-701

**B. TECH. (CIVIL ENGINEERING) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

DESIGN OF STEEL STRUCTURE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

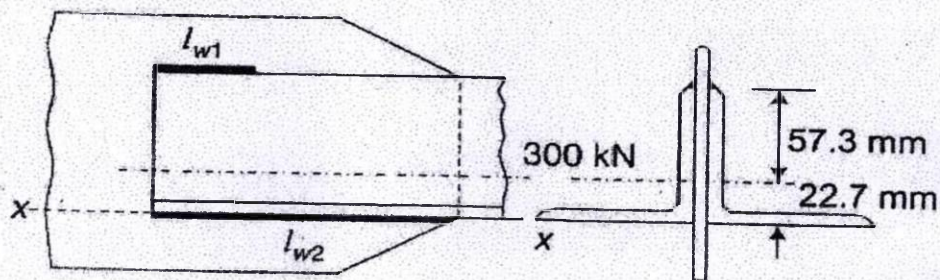
(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

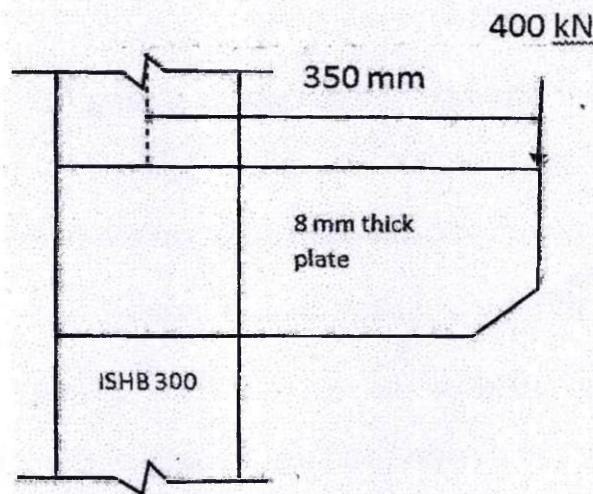
1. (a) A groove weld is to connect two plates $180 \text{ mm} \times 18 \text{ mm}$ each. Determine the design bending strength of the joint, if it is subjected to a moment of 13 kN-m . Also, determine the adequacy of the adequacy of the joint if the shear force at the joint is 200 kN . Assume the welds to be of double-U shop welded. (CO1)
- (b) A tie member of a truss consists of double angle section, each $80 \text{ mm} \times 80 \text{ mm} \times 8 \text{ mm}$ welded on the opposite side of a 12 mm thick gusset plate as shown in Fig. Design a fillet weld for making the

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connection in the workshop. The factored tensile force in the member is 300 kN. (Fe 410 grade steel for both connection and members). (CO1)



- (c) A bracket is bolted to the flange of a column as shown below using 8 mm thick plate. Using M20 bolts of grade 4.6. Design the connection for $f_u = 410$ MPa, $f_{ub} = 400$ MPa, $f_y = 250$ MPa, $f_{yb} = 240$ MPa. (CO1)

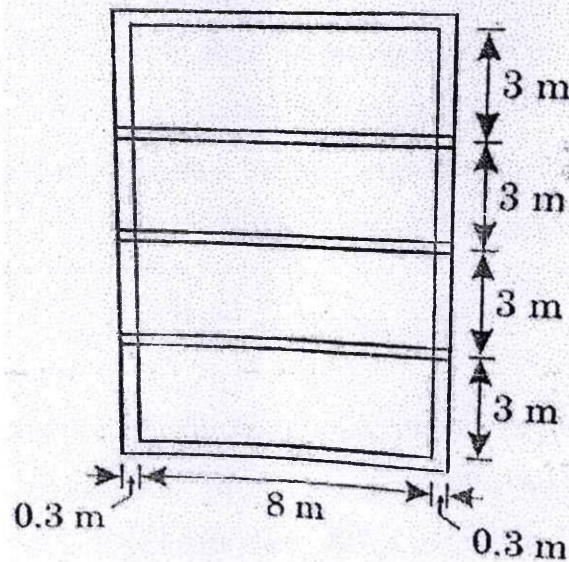


(All dimensions are in mm)

2. (a) Design a bridge truss diagonal subjected to a factored tensile load of 300 kN. The length of the diagonal is 3.0 m. The tension member is connected to a gusset plate 16 mm thick with one line of 20 mm diameter bolts of grade 8.8. (CO2)

- (b) Design a splice for joining tension member sections 160×10 mm and 250×14 mm. The member is subjected to a factored tensile load of 300 kN. Assume Fe 410 grade of steel. Provide 20 mm diameter bolts of grade 4.6 for making the connections. (CO2)
- (c) With the help of neat diagram write a short note on : (CO2)
- (i) Lug Angles
 - (ii) Splices
 - (iii) Gusset plates
3. (a) Design a stanchion consisting of two channel section for a working load of 750 kN. The effective length of column is 10 mtr. The section available are ISMC 250, ISMC 300 and ISMC 350. (CO3)
- (b) Design the lacing for connecting the two members of lacing and draw the suitable diagram for the column chosen above. (CO3)
- (c) An ISA is $100 \times 100 \times 6$ mm is connected to Gusset plate in a truss. The length of strut between the intersection at each end is 4.5 mtrs. Calculate the strength of strut if it is connected by 3 bolts at each end. Treat the end to be fixed-fixed. (CO3)
4. (a) A simply supported steel joist of 4.0 m effective span is laterally supported throughout. It carries a total uniformly distributed load of 40 kN (inclusive of self-weight). Design an appropriate section using steel of grade Fe 410. (CO4)
- (b) Determine the design bending strength and the design shear strength of a laterally supported beam ISMB 300. (CO4)
- (c) A roof of a hall measuring $8 \text{ m} \times 12 \text{ m}$ consists of 100 mm thick RCC slab supported on steel I-beams spaced 3 m apart as shown in figure

below. The finishing load may be taken as 1.5 kN/m^2 . Design the steel beam. (CO4)



5. (a) Design a strut in a roof truss for the following data : (CO5)
- Length of the strut = 2.235 m
 - Factored compressive force = 50 kN (due to D.L. and L.L.)
 - Factored tensile force = 17.80 kN (due to D.L. and W.L.)
 - Grade of steel Fe 410
 - Grade of bolts 4.6
 - Bolt diameter 20 mm.
- (b) Draw a roof truss and mention various components of roof truss. Draw different types of roof trusses. (CO5)
- (c) Describe the selection of roof truss and derive the expression for finding the economical spacing of roof trusses. (CO5)

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TCE-702

B. TECH. (CIVIL ENGG.) (SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

CONSTRUCTION MANAGEMENT AND PLANNING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

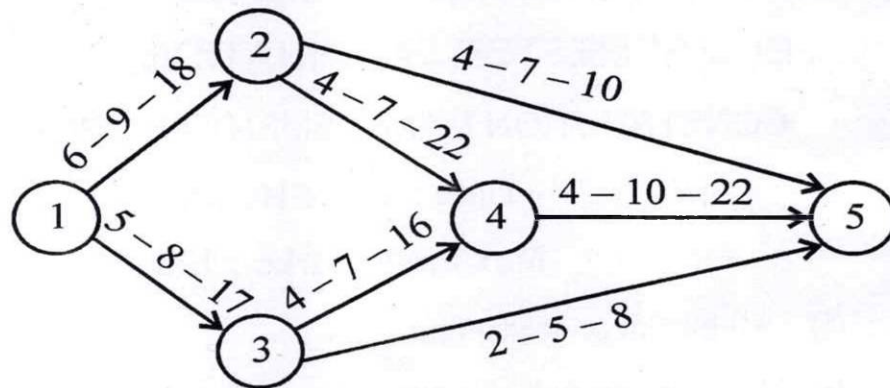
(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

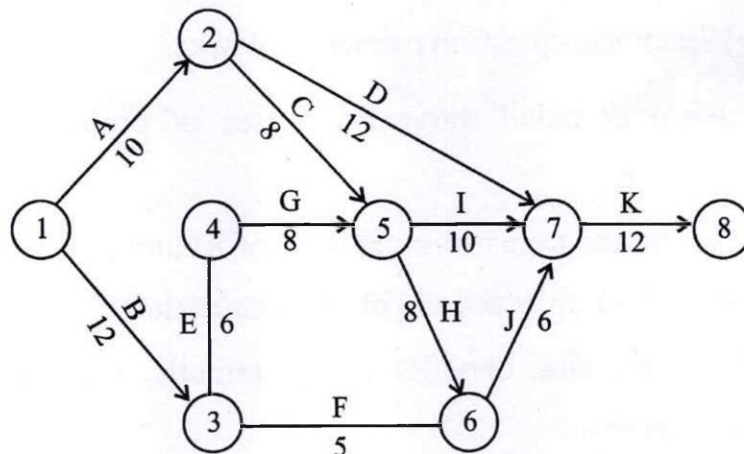
1. (a) Explain in detail the main causes of project failure in construction industries.
(b) Describe the terms Internal Rate of Return and Benefit-Cost Ratio. Also discuss its importance in civil construction industries.
(c) What are the benefits of systematic risk identification and risk management ?
2. (a) Explain steps involved in project planning and also describe the importance of project planning in civil construction works.

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- (b) Explain the role and impact of project management organization and staffing in civil construction works.
- (c) Explain in detail various categories of construction projects.
3. (a) The network of certain project shown in Fig. Determine the expected time for each of the path and also show the critical path :



- (b) For the given network as shown in Fig, compute : (i) event time, (ii) total float, (iii) free float and also determine the critical path :



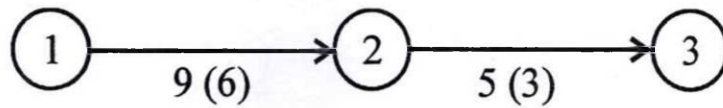
- (c) The following table gives the activities in a construction project and other relevant information :
- (i) Draw the network for the project.

(ii) Find the critical path.

(iii) Find free float and independent float for each activity.

Activity	Duration (Days)
1-2	20
1-3	25
2-3	10
2-4	12
3-4	6
4-5	10

4. (a) The normal, crash cost and time data of activities in network are given shown in the fig. and table. Determine the optimum project duration if project overhead costs are @ ₹ 850 per day :



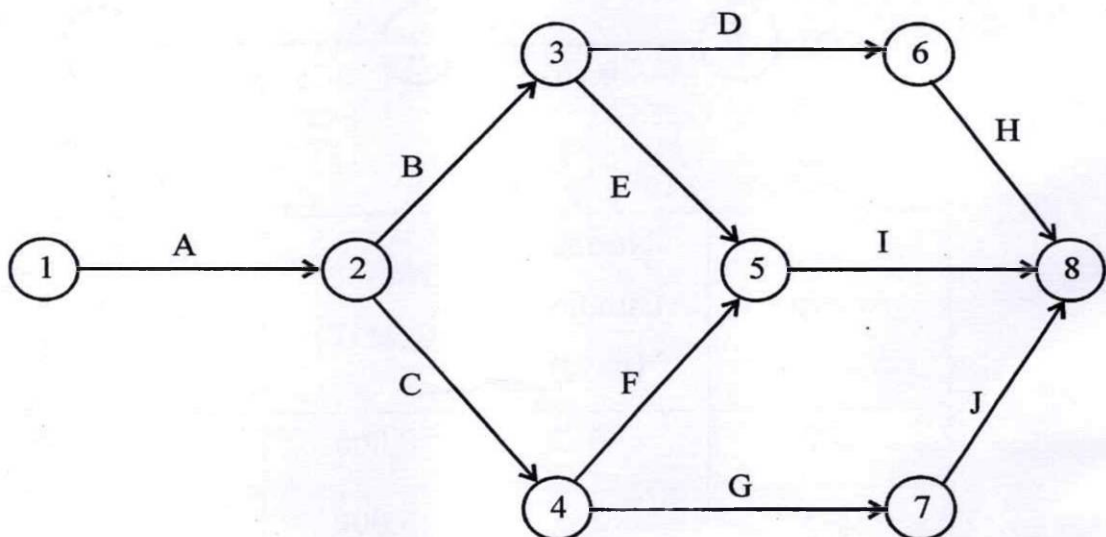
Activity	Normal Duration (days)	Normal Cost (₹)	Crash Duration (days)	Crash Cost (₹)
1-2	9	8,000	6	9,500
2-3	5	5,000	3	5,500

(b) Describe updating of project networks and resources allocation.

(c)

Activity	Estimated Duration	Activity	Estimated Duration
A	5	F	2
B	2	G	3
C	6	H	8
D	4	I	7
E	4	J	2

A building project consists of 10 activities represented by network shown in the given network diagram, and estimated time for each activity shown in table. Compute (i) event time, (ii) activity times and (iii) total float. Also determine critical path :



5. (a) Discuss various types of contract system with the relative merits and demerits.
- (b) Explain in detail all the items included in the tender document.
- (c) Explain the safety measures adopt in road construction works.



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TCE-703

**B. TECH. (CIVIL ENGINEERING) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

TRANSPORTATION ENGINEERING-II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the possible causes of creep ? What are the effects of creep ?

(CO1)

(b) Write short on the following :

(CO1)

(i) Wind resistance

(ii) Resistance due to track profile

(iii) Starting and accelerating resistance

(iv) Train resistance

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- (c) A locomotive is designated by 4.62 is to haul train at a speed of 75 km/h, load of each axle is 22 t, $\mu = 0.2$. Compute : (CO1)
- (i) What is the maximum load engine can pull on a straight and level track.
- (ii) When is to be run on profile slope 1 in 180, then the maximum speed it can take ?
2. (a) Mention the main function of ballast. If the spacing of sleepers is 1.2 metres and width of sleeper is 30 cm, find out the ballast cushion. (CO2)
- (b) Define composite sleeper index. Using a sleeper density of M+5, find out the number of sleepers required for constructing a railway track 640 metres long (B. G. Track). (CO2)
- (c) If theoretical length of long welded rail beyond which central portion of 52 kg/mtr. Rail would not be subjected to any longitudinal movement due to 35° temperature increase if : (CO2)

$$A_s = 66.15 \text{ cm}^2$$

$$\alpha = 11.5 \times 10^{-6} \text{ per } ^\circ\text{C}$$

$$E = 21 \times 10^6$$

$$S = 60 \text{ m}$$

$$F_{s1} = 300 \text{ kg}$$

$$\Delta t = 35^\circ$$

Calculate breathing length.

3. (a) How do you define the superelevation ? What are the objectives of providing superelevation on curves of a railway track ? (CO3)
- (b) Define grade compensation. If the ruling gradient is in 150 on a particular section of BG and the same time of curve of 4° is situated on this ruling gradient, what should be the allowable rolling gradient ? (CO3)
- (c) Determine the length of transition curve and drag the offsets at every 15 m. Given that the design speed of the train on curve is 90 kmph on a BG Track. (CO3)
4. (a) What are the objects of signalling ? Explain the different types of signals used in station yards. (CO4)
- (b) Explain the track junction. Enumerate the various types of track junction in use. (CO4)
- (c) For turn out on BG track, following properties are known : (CO4)
- No of crossing = 1 in 12
- Heel divergence $d = 12$ cm
- The curve is starting from toe of the switch and ends at TNC. Compute :
- (i) Curve lead (CL)
 - (ii) Radius (R)
 - (iii) Switch lead (SL)
 - (iv) Lead (L)
 - (v) Central radius

5. (a) What is the necessity of railway tunnels ? Draw a sketch to illustrate a single track railway tunnel. (CO5)
- (b) Name of the different methods of soft ground tunnelling and describe the American method in detail with diagram. (CO5)
- (c) Briefly explain drilling of shaft and drilling equipment. (CO5)

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TCE-712

**B. TECH. (CIVIL) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

GEOGRAPHIC INFORMATION SYSTEM

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the following in brief :

(i) What is GIS ?

(ii) What is the purpose of GIS ?

(iii) What types of data can be managed in GIS ?

(iv) How does GIS work ?

(v) What are the components of GIS ?

(b) Explain the following in brief :

(i) How is GIS used in urban planning ?

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- (ii) How is GIS be applied in environmental management ?
- (iii) What are the challenges in using GIS ?
- (iv) Can GIS be used for disaster management ?
- (c) Explain the following in brief :
 - (i) How does GIS benefits public health ?
 - (ii) What are some popular GIS software programs ?
 - (iii) How is GIS being used in agriculture ?
 - (iv) What is a geographic database in GIS ?
- 2. (a) Explain the concept of spatial data models in GIS. Provide examples of common spatial data models and their applications.
- (b) Describe common GIS data formats and their advantages and disadvantages.
- (c) Differentiate between attribute data and spatial data in GIS. Provide examples of each data type.
- 3. (a) Brief about Object Oriented Database and Database Management System.
- (b) Explain about Network Data Structure and Relational Database Structure.
- (c) Compare and contrast the advantages and disadvantages of using a file-based approach versus a database management system for GIS data storage.
- 4. (a) Explain what a buffer operation is in GIS and provide an exmaples of its practical application.
- (b) Describe the concept of a GIS overlay operation. Provide an example of how overlay analysis can be used to answer a real-world question.

(3)

- (c) Differentiate between raster-based and vector-based spatial analysis in GIS. Provide an example of when you might choose one approach over the other.
- 5. (a) Discuss the role of GIS in urban planning and administration. Provide example of how GIS can be used to address urban challenges.
- (b) Identify and explain each step involved in the process of site suitability analysis.
- (c) Develop flow chart justifying the statement 'GIS for site suitability in urban planning'.

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